

Somesh Suryawanshi

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CAREER OBJECTIVE

Mechanical Engineer combining physical Tool & Die manufacturing expertise with advanced R&D Simulation-Led Design (Ansys CFD/FEA), G2 curvature surface modeling, and GD&T. Published patent inventor and developer of automated Python-based CAD pipelines. Leverages AI-assisted workflows and edge computing to accelerate mechanical design, optimize fluid dynamics, and deploy production-ready digital twins.

EDUCATION

Marathwada Mitra Mandal's College of Engineering, Pune Expected Graduation: 2027
Bachelor of Mechanical Engineering (BE)

Indo German Tool Room, Sambhaji Nagar Completed
Advance Diploma in Tool & Die Making
• Distinction: 78.54%

PATENTS & PUBLICATIONS

IntelliPipe: Edge-Based Pipeline Monitoring & Anomaly Detection Published: March 2026
Lead Inventor | Patent Publication No: 202621020281 A

- Engineered an edge-computing IIoT monitoring system featuring an automated Python-based firmware factory that generates optimized C++ code for resource-constrained microcontrollers (ESP32).
- Deployed an in-memory digital twin performing real-time statistical self-calibration for autonomous leak and blockage detection without cloud dependency.

EXPERIENCE

Indo German Tool Room 2023 – 2024

Manufacturing Engineering Intern (Technical Sales & Floor Coordination)

- Directed technical client relations for 15+ manufacturing accounts, translating customer requirements directly into CAD design modifications and actionable shop-floor manufacturing specifications.
- Coordinated shop-floor assembly and fabrication for high-precision press tools, including shim punching tools, blanking dies, and multi-cavity hand moulds.
- Executed DFM/DFA principles and ASME Y14.5 GD&T standards to ensure final manufactured tools achieved strict alignment with initial client tolerances and design intent.

PROJECTS

Gaitform: Automated 3D Biomechanical CAD Pipeline

- Engineered a Python-based CAD pipeline (OpenCV, Trimesh, Streamlit) translating 1,260-node spatial pressure CSV data into custom 3D-printable orthotic shells maintaining a strict +/- 1.5 mm dimensional tolerance.
- Integrated a mathematical Gaussian smoothing algorithm to eliminate surface stress discontinuities in STL outputs prior to additive manufacturing.

Aerodynamic CFD Study: High-Speed Automotive Optimization

- Executed an end-to-end fluid dynamics simulation in Ansys Fluent utilizing an SST k-omega turbulence model and a 98k polyhedral mesh across a 140 km/h flow domain.
- Achieved full solver convergence at 172 iterations, optimizing vehicle surface curvature to reduce flow separation, yielding a drag coefficient (Cd) of 0.425 and lift coefficient (Cl) of -0.533.

CERTIFICATIONS & ACHIEVEMENTS

- Advance C++ Training, IIT Bombay Spoken Tutorial (Achieved 85.00% in proctored exam).
- Entrepreneurship & Skill Development in Tool/Product Analysis, ANSYS-MSME Technology Centre.
- Bike Crank Design Tutorial with Ansys Discovery, Ansys STEM Verified Achievement.

SKILLS

- CAE & Simulation:** Ansys Fluent (CFD), Ansys Mechanical (FEA), Structural Analysis, SST k-omega Turbulence.
- CAD & Surface Modeling:** SolidWorks (G2 Curvature Continuity), CATIA V5, Siemens NX CAD, AutoCAD.
- Manufacturing & Tolerancing:** GD&T (ASME Y14.5), Tool & Die Design, DFM/DFA, Sheet Metal.
- Automation & Programming:** Python (OpenCV, Trimesh, Streamlit), C/C++ Firmware Generation, Git/GitHub, Edge Computing.